



A.K.T.U – B.Tech First Year Syllabus

Content

Unit-1: Matrices

✓ completed

Elementary transformations, Inverse of a matrix, Rank of matrix, Solution of system of linear equations, Characteristic equation, Cayley-Hamilton Theorem and its application, Linear Dependence and Independence of vectors, Eigen values and Eigen vectors, Complex Matrices, Hermitian, Skew-Hermitian and Unitary Matrices, Applications to Engineering problems.

Unit-2: Differential Calculus- I

→ Easy

Successive Differentiation (nth order derivatives), Leibnitz theorem, Curve tracing, Partial derivatives, Euler's Theorem for homogeneous functions, Total derivative, Change of variables.

Unit-3: Differential Calculus-II

Expansion of functions by Taylor's and Maclaurin's theorems for functions of one and two variables, Maxima and Minima of functions of several variables, Lagrange's method of multipliers, Jacobians, Approximation of errors.



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Unit-4: Multiple integration

Double integral, Triple integral, Change of order of integration, Change of variables, Beta and Gamma function and their properties, Dirichlet's integral and its applications to area and volume, Liouville's extensions of Dirichlet's integral.

Unit-5: Vector Calculus

Vector differentiation: Gradient, Curl and Divergence and their Physical interpretation, Directional derivatives.

Vector Integration: Line integral, Surface integral, Volume integral, Gauss's Divergence theorem, Green's theorem and Stoke's theorem (without proof) and their applications.

UNIT –02

Differential Calculus– I



TOPICS *to be covered*

- ❖ ✓ Successive Differentiation (nth order derivatives). *calculation*
- ❖ ✓ Leibnitz theorem. *20 marks*
- ❖ ✓ Curve tracing. *EASY*
- ❖ ✓ Partial derivatives. *EASY*
- ❖ Euler's Theorem for homogeneous functions.
- ❖ Total derivative.
- ❖ Change of variables.





Successive Differentiation



Let

$$y = f(x)$$

y_1

1st derivative

$$\frac{dy}{dx} = \frac{d}{dx} F(x)$$

2nd derivative

$$\frac{d^2y}{dx^2} = \frac{d^2}{dx^2} F(x)$$

y_2

$$\frac{d^ny}{dx^n} = \frac{d^n}{dx^n} F(x)$$

nth derivative = y_n



Question

Find $\frac{d^3y}{dx^3}$, $y = 4x^3 + 4x + 2$.

$$y = 4x^3 + 4x + 2$$

$$\frac{dy}{dx} = y_1 = \frac{d}{dx}(4x^3 + 4x + 2)$$

$$y_1 = 12x^2 + 4 + 0$$

$$\frac{d^3y}{dx^3} = 24$$

$$\frac{dy_1}{dx} = y_2 = \frac{d}{dx}(12x^2 + 4)$$

$$y_2 = 24x + 0$$

$$\frac{dy_2}{dx} = y_3 = \frac{d}{dx}(24x)$$

$$y_3 = 24 \text{ Ans.}$$



Question

$y = a \cos(\log x) + b \sin(\log x)$, Prove that, $x^2 y_2 + x y_1 + y = 0$

$$\left(x \frac{dy_1}{dx} + y_1 \right) = -a \cos(\log x) \frac{1}{x} - b \sin(\log x) \frac{1}{x}$$

$\downarrow y_2$

$$x y_2 + y_1 = \frac{-a \cos(\log x) - b \sin(\log x)}{x}$$

$$x^2 y_2 + x y_1 = - (a \cos(\log x) + b \sin(\log x))$$

$$x^2 y_2 + x y_1 = -y$$

$$x^2 y_2 + x y_1 + y = 0$$

2nd derivative

$$\frac{d}{dx}(x y_1) = \frac{d}{dx}(-a \sin(\log x) + b \cos(\log x))$$

Prove that -

$$x^n y_n + x^{n-1} y_{n-1} + y = 0$$

$\downarrow$
nth-derivative.



Calculation of n^{th} Order Derivatives

I. To find the n^{th} differential co-efficient of x^m .

Let

$$y = x^m$$

$$\frac{d^m}{dx^m} (x^m) = m!$$

$$\rightarrow y = x^{\textcircled{2}}$$

$$\frac{d^2 y}{dx^2} = 2! = 2 \times 1 = \textcircled{2}$$

II. The n^{th} differential co-efficient of $(ax + b)^m$, where $n < m$.

Let

$$y = (ax + b)^m$$

$$\therefore y_n = m(m-1)(m-2)(m-3)\dots(m-n+1)(ax+b)^{m-n} \cdot a^n$$



Calculation of n^{th} Order Derivatives

III. To find the n^{th} differential co-efficient of $\frac{1}{ax+b} \left[x \neq -\frac{b}{a} \right]$

Let

$$y = \frac{1}{ax+b} = (ax+b)^{-1}$$

Or

$$y_n = \frac{(-1)^n n! a^n}{(ax+b)^{n+1}}$$

IV. The n^{th} differential co-efficient of $\log(ax+b)$.

Let

$$y = \log(ax+b)$$

or

$$y_n = \frac{(-1)^{n-1} (n-1)! a^n}{(ax+b)^n}$$

$$\begin{aligned} & \log x + \log(a+x) \\ & \downarrow \\ & \log(1x+0) + \log(1x+a) \\ & \quad \downarrow \quad \quad \downarrow \\ & \quad a=1 \quad b=0 \quad \quad a=1 \quad b=a \end{aligned}$$



Calculation of n^{th} Order Derivatives

V. The n^{th} differential co-efficient of a^{mx}

Let

$$y = a^{mx}$$

$\therefore$

$$y_n = m^n \cdot a^{mx} (\log a)^n$$

VI. The n^{th} differential co-efficient of $\sin(ax + b)$

Let


$$y = \sin(ax + b)$$

$$y_n = a^n \sin\left(ax + b + n\frac{\pi}{2}\right)$$



Calculation of n^{th} Order Derivatives

VII. The n^{th} differential co-efficient of $\cos (ax + b)$

Let

$$y = \cos (ax + b)$$

$$y_n = a^n \cos \left(ax + b + n \frac{\pi}{2} \right)$$

VIII. The n^{th} differential co-efficient of $e^{ax} \sin (bx + c)$

Let

$$y = e^{ax} \sin (bx + c)$$

$$y_n = (a^2 + b^2)^{n/2} e^{ax} \cdot \sin \left(bx + c + n \cdot \tan^{-1} \frac{b}{a} \right)$$



Calculation of n^{th} Order Derivatives

IX. The n^{th} differential co-efficient of $e^{ax} \cos (bx + c)$

Let

$$y = e^{ax} \cos (bx + c)$$

$$y_n = (a^2 + b^2)^{n/2} e^{ax} \cdot \cos \left(bx + c + n \cdot \tan^{-1} \frac{b}{a} \right)$$



Question

$$y_n = (-1)^{n-1} (n-1)! \left[\frac{1}{x^n} + \frac{1}{(x+a)^n} \right]$$

Find n^{th} differential coefficient of $\log(ax + x^2)$.

$$\text{Let } y = \log(ax + x^2)$$

$$y = \log x \cdot (a+x)$$

$$y = \log x + \log(a+x)$$

$$\# \log(ax+b) \rightarrow y_n = \frac{(-1)^{n-1} (n-1)! a^n}{(ax+b)^n}$$

$$y = \log(1x+0) + \log(1x+a)$$

$$\frac{d^n y}{dx^n} = \frac{d^n}{dx^n} \log(1x+0) + \frac{d^n}{dx^n} \log(1x+a)$$

$a=1, b=0$ $a=1, b=a$

$$y_n = \frac{(-1)^{n-1} (n-1)! \overset{1}{\circlearrowleft} 1^n}{x^n} + \frac{(-1)^{n-1} (n-1)! \overset{1}{\circlearrowleft} 1^n}{(x+a)^n}$$

$$y_n = \frac{(-1)^{n-1} (n-1)!}{x^n} + \frac{(-1)^{n-1} (n-1)!}{(x+a)^n}$$



Question

If $y = \sin 2x \sin 3x$. Find n^{th} differential coefficient.

$$y = \frac{1}{2} (2 \sin 2x \sin 3x)$$

$$y = \frac{1}{2} (\cos x - \cos 5x)$$

$$y = \frac{1}{2} \cos x - \frac{1}{2} \cos 5x$$

Diff. n -times

$$y_n = \frac{1}{2} \frac{d}{dx^n} \cos(x+0) - \frac{1}{2} \frac{d}{dx^n} \cos(5x+0)$$

$a=1$ $a=5$
 $b=0$ $b=0$

$$y = \cos(ax+b) \longrightarrow y_n = a^n \cos(ax+b+n\pi/2)$$

$$y_n = \frac{1}{2} \left(1^n \cos(x+n\pi/2) \right) - \frac{1}{2} \left(5^n \cos(5x+n\pi/2) \right)$$

$$y_n = \frac{\cos(x+n\pi/2)}{2} - \frac{5^n \cos(5x+n\pi/2)}{2}$$



Question

Find n^{th} derivative of,

$$y = \frac{1}{2a} \left(\frac{1}{x-a} - \frac{1}{x+a} \right)$$

Diff. n -times -

$$y_n = \frac{1}{2a} \left[\frac{d^n}{dx^n} \frac{1}{x-a} - \frac{d^n}{dx^n} \frac{1}{x+a} \right]$$

$$\therefore \frac{1}{2a} \left[\frac{(-1)^n n!}{(x-a)^{n+1}} - \frac{(-1)^n n!}{(x+a)^{n+1}} \right]$$

$$y_n = \frac{(-1)^n n!}{2a} \left[\frac{1}{(x-a)^{n+1}} - \frac{1}{(x+a)^{n+1}} \right] \quad \text{Ans}$$

$$\frac{1}{ax+b}$$

$$\frac{(-1)^n n! a^n}{(ax+b)^{n+1}}$$

(i) $\frac{1}{x^2-a^2}$

(ii) $\frac{2x+1}{(2x-1)(2x+3)}$

$$y = \frac{1}{2} \left[\frac{1}{2x-1} + \frac{1}{2x+3} \right]$$

$$\begin{matrix} a=2 \\ b=-1 \end{matrix} \quad \begin{matrix} a=2 \\ b=3 \end{matrix}$$

[U.P.T.U : 2015]

$$\frac{2x+1}{(2x-1)(2x+3)} = \frac{A}{2x-1} + \frac{B}{2x+3}$$

$$A \left| \begin{matrix} 2x-1=0 \\ x=1/2 \end{matrix} \right. = \frac{2}{4} = 1/2$$

$$B \left| \begin{matrix} 2x+3=0 \\ x=-3/2 \end{matrix} \right. = \frac{-2}{-4} = 1/2$$